

TYLER ROBERTS

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 9+ years of experience designing, building, and operating cloud-native, AI-enabled systems with a strong focus on machine learning, NLP, and Generative AI. Proven expertise developing end-to-end ML solutions including data preprocessing, feature engineering, model training, validation, deployment, and monitoring in production environments. Hands-on experience with Python-based ML stacks, LLMs, RAG pipelines, vector databases, and agentic workflows. Strong background deploying and operating ML workloads on Azure, GCP, and AWS using MLOps best practices. Adept at collaborating cross-functionally with data engineers, product teams, and stakeholders to translate complex business problems into scalable, reliable ML solutions.

SKILLS

Python	JavaScript	TypeScript	Node.js
Scikit-learn	TensorFlow	PyTorch	NumPy
Pandas	Natural Language Processing (NLP)	Large Language Models (LLMs)	Generative AI
Prompt Engineering	Retrieval-Augmented Generation (RAG)	LangChain	Hugging Face
MLflow	Kubeflow	Azure ML Studio	Vertex AI
Model Monitoring	Automated ML Pipelines	Azure	Google Cloud Platform (GCP)
AWS	Cloud-Native Architecture	Feature Engineering	Data Preprocessing
Model Evaluation	Vector Databases	Docker	CI/CD
Infrastructure as Code	Observability		

EXPERIENCE

Senior Software Engineer

Web Venturez LLC

February 2020 - October 2025

- Designed and implemented end-to-end machine learning and AI-driven features, including NLP pipelines, LLM integrations, and predictive analytics systems deployed to production environments.
- Built scalable data preprocessing and feature engineering pipelines using Python, NumPy, Pandas, and Scikit-learn to support model training and inference workflows.
- Developed and deployed ML models to cloud platforms including Azure and GCP, leveraging managed services and containerized deployments for reliability and scalability.
- Implemented MLOps practices such as automated training pipelines, experiment tracking, model versioning, and performance monitoring using tools like MLflow and cloud-native ML platforms.
- Collaborated closely with data engineers, analysts, and product stakeholders to translate business requirements into robust ML solutions with clear performance metrics and validation strategies.

Full Stack Engineer

Webiz Digital

July 2017 - January 2020

- Developed data-driven web applications and backend services that supported analytics, reporting, and early-stage machine learning use cases.
- Built RESTful APIs and backend workflows in Node.js and Python to support data ingestion, transformation, and downstream analytical processing.
- Collaborated with cross-functional teams to integrate data pipelines and analytics features into customer-facing applications.
- Optimized application performance and data handling through efficient query design, caching strategies, and structured data models.
- Supported cloud deployments and CI/CD pipelines, gaining hands-on experience with AWS-based infrastructure and automation.
- Contributed to improving code quality and maintainability through reviews, refactoring, and adoption of best practices.

Software Developer

Centric Tech Inc.

October 2013 - June 2017

- Built and maintained backend services and internal tools supporting data-heavy business workflows and reporting systems.
- Implemented application logic and APIs using JavaScript and backend technologies, contributing to early distributed system designs.
- Worked closely with senior engineers to understand system architecture, data flows, and production deployment processes.
- Assisted with debugging, performance tuning, and reliability improvements in production environments.
- Collaborated across teams to deliver features aligned with business requirements and timelines.
- Developed a strong foundation in software engineering, data handling, and system reliability that later supported advanced ML and AI work.

EDUCATION

Bachelor of Computer Science

Florida State University

2009 - 2013